Austin Brown | C.V.

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Research interests

My main research focus is on the design, reliability, and computational efficiency of Markov chain Monte Carlo algorithms by studying their convergence properties with optimal transportation. Creating reliable, computationally efficient Markov chain Monte Carlo algorithms in high dimensions for practitioners is a central motivation for my research. Additionally, I am interested in applications of Markov chain Monte Carlo to Bayesian error-in-variable models for machine learning, astrophysics, and epidemiology. My other research interests are in applications of optimal transport to stochastic processes, and I am also interested in combining optimal transport and theory for Markov chain Monte Carlo to study generative A.I. models in the future.

My dissertation "Some Convergence Results for Metropolis-Hastings Algorithms" is available online `https://conservancy.umn.edu/handle/11299/243073`.

Education

University of Minnesota - Twin Cities **Minneapolis, MN**
Ph.D. in Statistics *August 2022*
Dissertation: "Some Convergence Results for Metropolis-Hastings Algorithms"
Advisor: Galin L. Jones
Committee: Charles Geyer, Qian Qin, Wei-Kuo Chen

University of Florida **Gainesville, FL**
B.S. in Mathematics *July 2017*
Major in Mathematics and minor in Statistics

Appointments

University of Toronto - St George **Toronto, Ontario**
Postdoctoral Fellow *July 2023–July 2025*
Advisor: Jeffrey S. Rosenthal

University of Warwick **Coventry, England**
Research Fellow *September 2022–June 2023*
Advisor: Krzysztof Łatuszynski

Publications and preprints

[1] A. Brown and J. S. Rosenthal, "Upper and lower bounds on the subgeometric convergence of adaptive Markov chain Monte Carlo," *Preprint, submitted*, 2024.

[2] A. Brown and J. S. Rosenthal, "Weak convergence of adaptive Markov chain Monte Carlo," *minor revision requested, Journal of Applied Probability*, 2024.

[3] S. Sixta, J. S. Rosenthal, and A. Brown, "Bounding and estimating MCMC convergence rates using common random number simulations," *Preprint, in peer review at Stochastic Models*, 2024.

[4] A. Brown, "A non-asymptotic error analysis for parallel Monte Carlo estimation from many short Markov chains," *Preprint, In peer review at Statistics and Computing*, 2024.

[5] A. Brown, K. Łatuszyński, and G. Roberts, "Adaptive pseudo-marginal Metropolis-Hastings," *In progress*, 2023.

[6] A. Brown and G. L. Jones, "Exact convergence analysis for Metropolis–Hastings independence samplers in Wasserstein distances," *Journal of Applied Probability*, vol. 61, no. 1, p. 33–54, 2024.

[7] A. Brown and G. L. Jones, "Lower bounds on the rate of convergence for accept-reject-based Markov chains in Wasserstein and total variation distances," *Bernoulli, to appear*, 2025.

[8] A. Brown, "Geometric ergodicity of Gibbs samplers for Bayesian error-in-variable regression," *Electronic Journal of Statistics*, vol. 18, no. 1, pp. 1495–1516, 2024.

[9] A. Brown and G. L. Jones, "Convergence rates of Metropolis–Hastings algorithms," *WIREs Computational Statistics*, vol. 16, no. 5, p. e70002, 2024.

Seminar talks and conference posters

University of Warwick **Coventry, United Kingdom**
Algorithms and Computationally Intensive Inference Seminar *January 2023*
Talk Title: Lower Bounds on the Rate of Convergence for Accept-Reject-Based Markov Chains
Slides: `https://austindavidbrown.github.io/talk/warwick2023.pdf`

University of Warwick **Wales, United Kingdom**
University of Warwick Departmental Conference 2023 *April 2023*
Talk Title: Exact convergence for independence samplers in Wasserstein distance.
Slides: `https://austindavidbrown.github.io/talk/warwickdepartmental2023.pdf`

University of Oxford **England, United Kingdom**
Bioinference 2023 *June 2023*
Poster Title: Geometric ergodicity of Gibbs samplers for Bayesian error-in-variable regression
Poster: `https://austindavidbrown.github.io/poster/bioinference2023.pdf`

University of Florida **Gainesville, FL**
University of Florida Statistics Seminar *September 2023*
Talk Title: Lower Bounds for Metropolis-Hastings in Wasserstein distances
Slides: `https://austindavidbrown.github.io/talk/uf2023.pdf`

University of Toronto **Toronto, Canada**
University of Toronto B.B. Departmental Seminar *January 2024*
Talk Title: Error analysis for parallel Monte Carlo estimation from many short Markov chains
Slides: `https://austindavidbrown.github.io/talk/toronto2024.pdf`

Teaching experience

University of Minnesota - Twin Cities **Minneapolis, MN**
Instructor ($\sim$ 70 students) *Spring 2020*
Regression and Correlated Data (STAT 3032)

University of Minnesota - Twin Cities **Minneapolis, MN**
Instructor ($\sim$ 80 students) *Spring 2021*
Introduction to Statistical Analysis (STAT 3011)

University of Toronto - St. George **Toronto, Canada**
Instructor ($\sim$ 250 students) *Fall 2023*
Methods of Data Analysis 1 (STA 302)

University of Toronto - St. George **Toronto, Canada**
Instructor ($\sim$ 300 students) *Fall 2024*
Methods of Data Analysis 1 (STA 302)

Academic service

I have refereed 6 articles for statistics and applied probability journals including Bayesian Analysis, Journal of Computational and Graphical Statistics, PLOS ONE, Advances in Applied Probability, Journal of Applied Probability, and Sankhya, Series B. I have also assisted in editing *An Introduction to Envelopes* by Dennis Cook.

I have a have taken part in the following committees and organizations:

University of Warwick ACII Seminar (Organizer)	*Jan. 2023–June 2023*
University of Warwick Statistics Department IT Committee	*Sept. 2022–June 2023*
University of Warwick Stonewall Self-Assessment (LGBTQ+ Inclusion)	*Sept. 2022–June 2023*
American Statistical Association LGBTQ+ Advocacy Committee	*Aug. 2023–Aug. 2024*
University of Toronto Brown Bag Departmental Seminar (Co-organizer)	*Sept. 2023–Present*

Organizations and memberships

Institute of Mathematical Statistics	*2022–2023*

Funding, scholarships, and awards

Beverly and Richard Fink Fellowship ($56000+Tuition) *2017–2022*

Lynn Lin Fellowship ($2000) *2019–2019*

First Year Scholarship ($2500) *2017–2017*

Mentorships

University of Toronto - St George **Toronto, Canada**

2024

I have acted as a mentor to 2 PhD students and 1 undergraduate student on their own individual research projects

Extracurricular activities

- Markov Chain Monte Carlo Group (2019–2022)
  Discussion of the state of the art MCMC literature

Statistical software

cmhi: A Python package for the centered Metropolis-Hastings independence algorithm. github.com/austindavidbrown/Centered-Metropolis-Hastings

mhlb: A Python implementation to estimate lower bounds on the geometric convergence rate for RWM Metropolis-Hastings. https://github.com/austindavidbrown/lower-bounds-for-Metropolis-Hastings

Programming language proficiency

Python: Pytorch, Autograd, Numpy, Matplotlib *Expert*

R: Rmarkdown *Expert*

C++: C++14 specification and above *Intermediate*

Languages

- English (native)
- French (novice)
- Spanish (novice)

References

- Galin Jones. University of Minnesota - Twin Cities. Email: galin@umn.edu
- Jeffrey S. Rosenthal. University of Toronto. Email: jeff@math.toronto.edu
- Qian Qin. University of Minnesota - Twin Cities. Email: qqin@umn.edu

∘ Krzysztof Latuszynski. University of Warwick. Email: K.G.Latuszynski@warwick.ac.uk